Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/05/2004
PI Name:	Fischer, David G Ph.D.		
Project Title:	Three-dimensional, Reflection-Mode Near-Field Microscopy for Microfluidic Phenomena		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	10/01/2001	**End Date:**	10/01/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Carney, Scott (University of Illinois)		
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Task Description:	The microscope under development is based on a recently formulated imaging procedure known as total internal reflection tomography (TIRT). It involves illuminating a sample with evanescent waves of varying penetration depth (generated by total internal reflection at the exit face of a prism) and detecting the scattered field as a function of angle. The use of evanescent waves, having sub-wavelength transverse structure, allows one to probe sample features on that scale also. Evanescent waves decay exponentially with increasing distance from their point of generation, however, so one is limited to imaging object structure within a couple of wavelengths of the exit face of the prism. The data set generated (or synthesized) by this measurement scheme, i.e. evanescent-wave diversity with angularly resolved scattered field detection, contains three-dimensional sub-wavelength structural information that can be inverted, enabling object reconstruction. The most promising inversion algorithm is one based upon the singular value decomposition (SVD) of the linearized scattering kernel describing the scattering process. The SVD is a mode decomposition that allows for a natural regularization of the inverse problem. The regularization, which stabilizes the solution, is imperative when the data is corrupted by noise. The maximum spatial resolution achievable with this microscope is a factor of n smaller than the typical half-wavelength resolution limit, where n is the index of refraction of the prism. Thus, higher index prisms allow better resolution, and this is due to the fact that they generate waves with greater degrees of evanescence (i.e. shorter penetration depths or probe scale). The objective of the proposed research is the development of a novel near-field optical microscope capable of three-dimensional sub-wavelength imaging. This work is based on recent theoretical advances regarding the fundamental interaction of light with sub-wavelength structures, as well as the development of stable algorithms for the near-field inverse problem. The microscope will be unique in its functionality, and may significantly advance the study of certain phenomena in fluid physics, including interfacial phenomena, cellular systems, and biocrystal growth While x-ray crystallography and electron microscopy have pushed the fundamental limits of resolution to the atomic scale, many applications are not amenable to such techniques. The reasons for this are many including non-periodicity of the sample, the impracticality of making in vivo observations in vacuum, and the undesirability of the destructive nature of the techniques on biological systems. In addition, some samples present high contrast only in certain regions of the electromagnetic spectrum, dictating the use of a certain frequency of probe field offering limited resolution. For example, standard optical techniques are limited by diffraction to a resolution of half the illuminating wavelength. This resolution limit is not fundamental, but due to the fact that measurements of the optical field are typically made many wavelengths from the sample. Recently, there has been interest in so-called near-field techniques that utilize the sub-wavelength information carried by the evanescent (or non-propagating) waves. In fact, steps in this direction have been taken within the Microgravity program, where a near-field optical microscope was constructed to observe protein crystal growth. A key limitation in state of the art near-field microscopy, however, is that its functionality is limited to the inspection of surfaces or optically thin media, i.e. effectively two-dimensional media. The development of a technique to take advantage of the sub-wavelength resolution provided by near-field optical microscopy while simultaneously extending the functionality of the method to provide three-dimensional imaging will have a substantial impact in the Microgravity community, both on ground-based as well as space-based studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improved sub-micron three-dimensional optical imaging.
Task Progress:	We have made substantial progress during our third year of funding. We have incorporated a custom-made high-index (low scatter) glass prism into our prototype microscope for the generation of evanescent waves. We performed initial tests of our microscope using 0.56 micron polystyrene spheres as a reference target. For a fixed mode of evanescent illumination, we recorded holographically the complex scattered field (both the amplitude and phase) generated by the polystyrene spheres. We believe that these are the first scattered field holograms recorded using an evanescent illumination source. We are now in the process of optimizing the incident beam scanning module, so that a set of near-field holograms can be recorded (corresponding to different evanescent illumination fields), and three-dimensional structure determination can be made. In addition to the construction of the TIRT microscope, we have continued our theoretical and numerical work regarding the information content of the near-field and the effect of limited and noisy measurement data on the stability and quality of TIRT object reconstruction. This work has provided invaluable guidance in optimizing the three-dimensional reconstruction algorithms. Our work has resulted in a two peer-reviewed journal publications and a book chapter.
Bibliography Type:	Description: (Last Updated: 04/24/2008)
Articles in Peer-reviewed Journals	Frazin RA, Fischer DG, Carney PS. "Information content of the near field: two-dimensional samples." JOSA A (Journal of the Optical Society of America A) 2004 Jun;21(6):1050-7. PMID: 15191187 ; http://dx.doi.org/10.1364/JOSAA.21.001050 , Jun-2004
Articles in Peer-reviewed Journals	Frazin RA, Fischer DG, Carney PS. "Information content of the near field: three dimensional samples." JOSA A (in press), October 2004. , Oct-2004
Books/Book Chapters	Fischer DG, Carney PS. "Total Internal Reflection Tomography (TIRT) for Three-Dimensional Sub-Wavelength Imaging." in "Tribute to Emil Wolf: Science and Engineering Legacy of Physical Optics. (SPIE Press Monograph Vol. PM139)" Ed. T.P. Jannson. Bellingham, WA : S P I E-International Society for Optical Engineering, September 2004., Sep-2004